ISIT JUNE 22-27 ANN ARBOR MICHIGAN

Hackathon

where quantum computing meets information theory

Entanglers

The Entanglers

Quantum Beamforming for IRS

__Problem__: Beamforming Algorithm for IRS
__Solution__: VQAs with good initial conditions

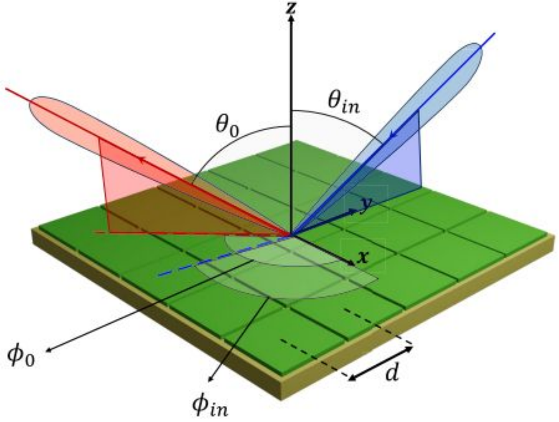
Quantum Annealing and QAOA

D-Wave and Pennylane

Faster than classical heuristics, while accounting for constraints

IRS Beamforming

$$G(\theta, \phi) = \frac{1}{MN} \sum_{m=1}^{M} \sum_{n=1}^{N} w_{m,n} e^{j\varphi_{m,n}(\theta,\phi)}$$



$$\max_{w_{m,n}} |G(\theta_0, \phi_0)|^2 \quad \text{s.t. } w_{m,n} \in \{1, -1\} \quad \forall m, n$$

Constraint:

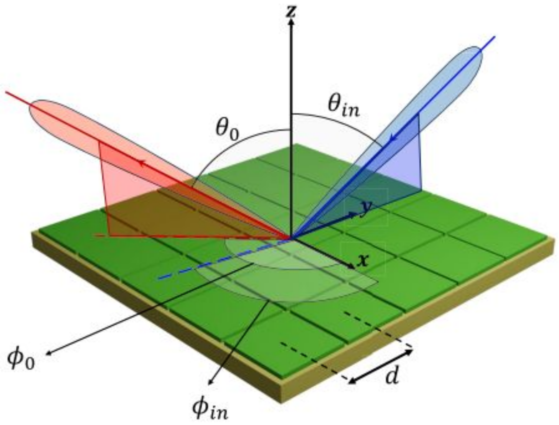
$$\text{subject to: } |G(\theta_i, \phi_i))| < c_i \quad \forall\, i \in I$$

[1]

IRS Beamforming

SSN Algorithm [2]

- When there are no constraints
- Provably optimal
- Linear time



Classical Failures! ⚠️

- No extension of SSN for the constrained case *[2]*
- SDP does not yield good results for the constrained problem *[2]*
- Simulated annealing takes too long for QCQP of bigger size *[3]*

ISIT 2025 Quantum Hackathon :
Group - Entanglers

Vital Ingredients

$$\max_{x \in \{-1,1\}^{MN}} x^T Q_0 x - \sum_i \lambda_i x^T Q_i x$$

QAOA

- Optimized λ parameters for meeting constraints
- Warm-starting QAOA by using parameterized X-Y mixers *[4]*
- Using ADAM optimizer for quick convergence time

Quantum Annealing

- Optimized λ parameters for meeting constraints
- Reverse Annealing on D-Wave Ocean for best initial state

On to the Code Demo!

Impact

- ❖ Cutting-edge area redefining **wireless** communications
- ❖ Allows for higher cell density allowing **higher data rates**
- ❖ Ensures non-leakage, useful in **defence** settings

Business Sustainability & Scalability

- Quantum Algorithms can solve energy overuse crisis
- Constant time complexity gives benefit at large scales
- Can allow long distance secure 6G communication in near future

Next Steps

- ❑ Implement constrained QAOA via Quantum Alternating Operator Ansatz for hard constraints

Entanglers

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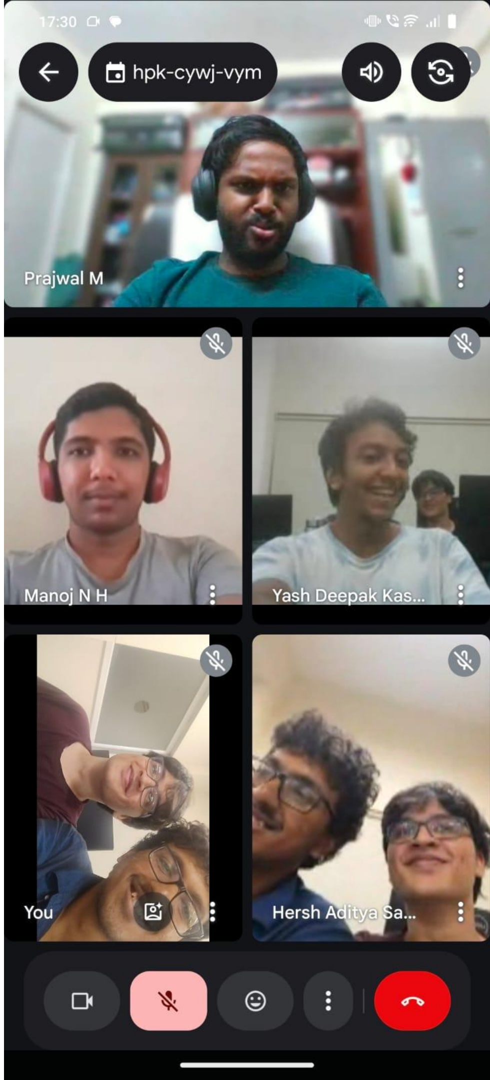

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References

1. S. S. Narayanan, R. K. Ganti, and U. K. Khankhoje, “Discrete Variable Beamforming for Intelligent Reflecting Surfaces,” in 2024 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio
Science Meeting (AP-S/URSI), 2024

2. SS Narayanan, R. K. Ganti, and U. K. Khankhoje, “Optimum Beamforming and Grating-Lobe Mitigation for Intelligent Reflecting Surfaces” in 2024 IEEE Transactions on

3. Das, A.; Chakrabarti, B. K. (2008). "Quantum Annealing and Analog Quantum Computation". *Rev. Mod. Phys.* **80** (3): 1061–1081. arXiv:0801.2193. Bibcode:2008RvMP...80.1061D. CiteSeerX 10.1.1.563.9990.